

CIPHER SHIFT

WC5-03: Next-Gen Length-Based Encryption

An innovative cryptographic experiment developed for the
ASSISI COMPETITION, blending modern aesthetics with
dynamic security logic.

Project Overview

avenchaos43654-droid / WC5-03

| THE INNOVATION

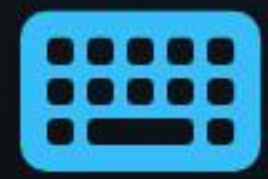
Length-Based Logic

Unlike standard Caesar ciphers that rely on fixed shifts, WC5-03 introduces a **Dynamic Shift Factor**. The length of the first word in the payload determines the rotation depth for the entire message.

Interactive Feedback

Security is no longer a silent process. The UI utilizes real-time tracking and intensive visual "Zap" animations to indicate successful matrix exchanges and encryption status.

| HOW IT WORKS



Input

User provides a plaintext payload via the glassmorphism interface.



Processing

The Flask backend calculates word length and applies the modular shift.



Output

The "Exchange Matrix" zaps the screen, revealing the ciphertext result.

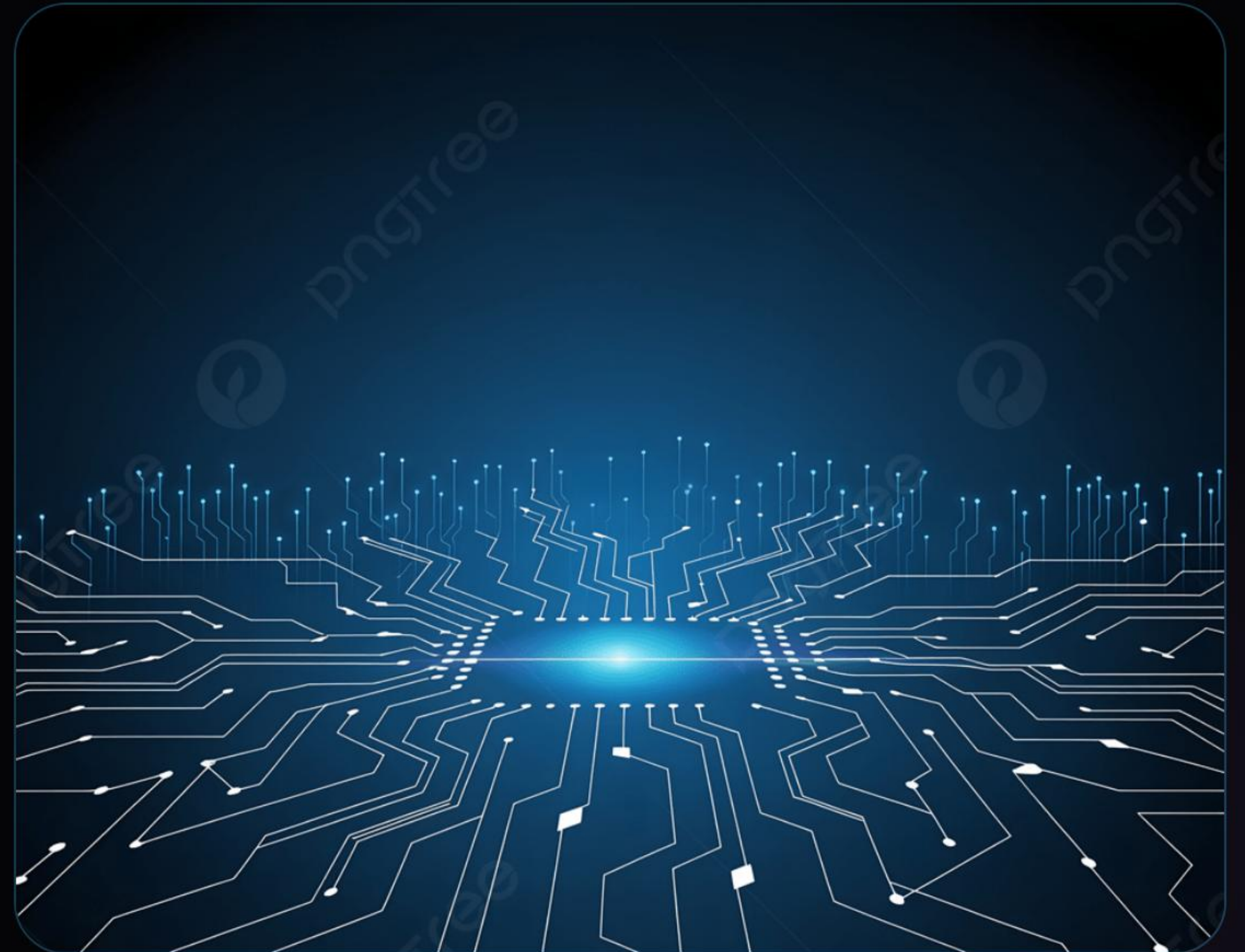
TECHNICAL ARCHITECTURE

Built with a robust, modular stack designed for speed and visual fidelity:

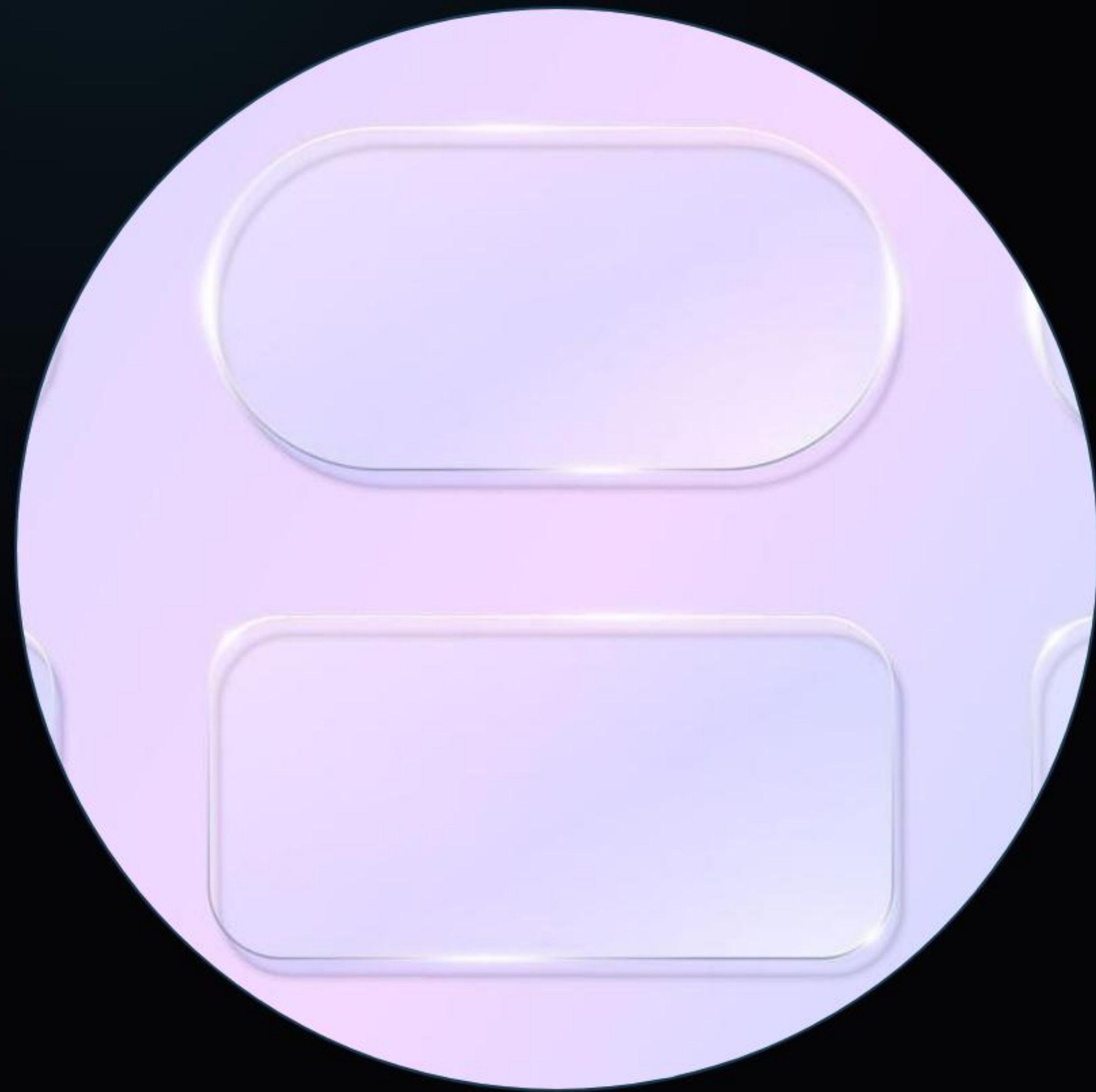
- ✔ **Backend:** Flask (Python)
- ✔ **Frontend:** Tailwind CSS & Lucide Icons
- ✔ **Animations:** Custom CSS @keyframes & JS Emitters
- ✔ **Logic:** Modular Modular Arithmetic Logic

Encryption Formula:

$$E_n(x) = (x + L) \bmod 26$$



| THE "VIBE" DESIGN



Aesthetic Philosophy

The project employs **Glassmorphism** to create a layered, immersive feel. Using high-intensity blurs and subtle borders, the UI mimics a frosted glass panel floating in a digital void.

Hyper-fluid gradients stream in the background, ensuring the application feels alive even when idle.

| USER EXPERIENCE

0ms

Cursor Latency

Immersive Realism

WC5-03 utilizes a **Zero-Delay Real-Time Cursor Tracking Engine**. The custom cursor isn't just a dot; it's a dynamic light source that interacts with the glass panels, creating localized glows and shadows as you move.

THE ZAP ANIMATION

Energy Feedback

When the "Exchange Matrix" protocol is triggered, the system initiates a high-intensity zap. The loading bar expands at supersonic speed while the core spinner "explodes" with neon particles.

This provides immediate psychological closure to the encryption process, making it feel powerful and definitive.

PROJECT STRUCTURE

Component	Path	Functionality
App Core	/app.py	Flask routes, POST processing, and cipher logic.
Templates	/templates/index.html	Jinja2 powered HTML with embedded Tailwinds.
Assets	/static/js	Cursor engine, particle emitter, and Zap logic.
Prompts	/snapshots/prompts	Documentation of development iterations.

POTENTIAL USE CASES



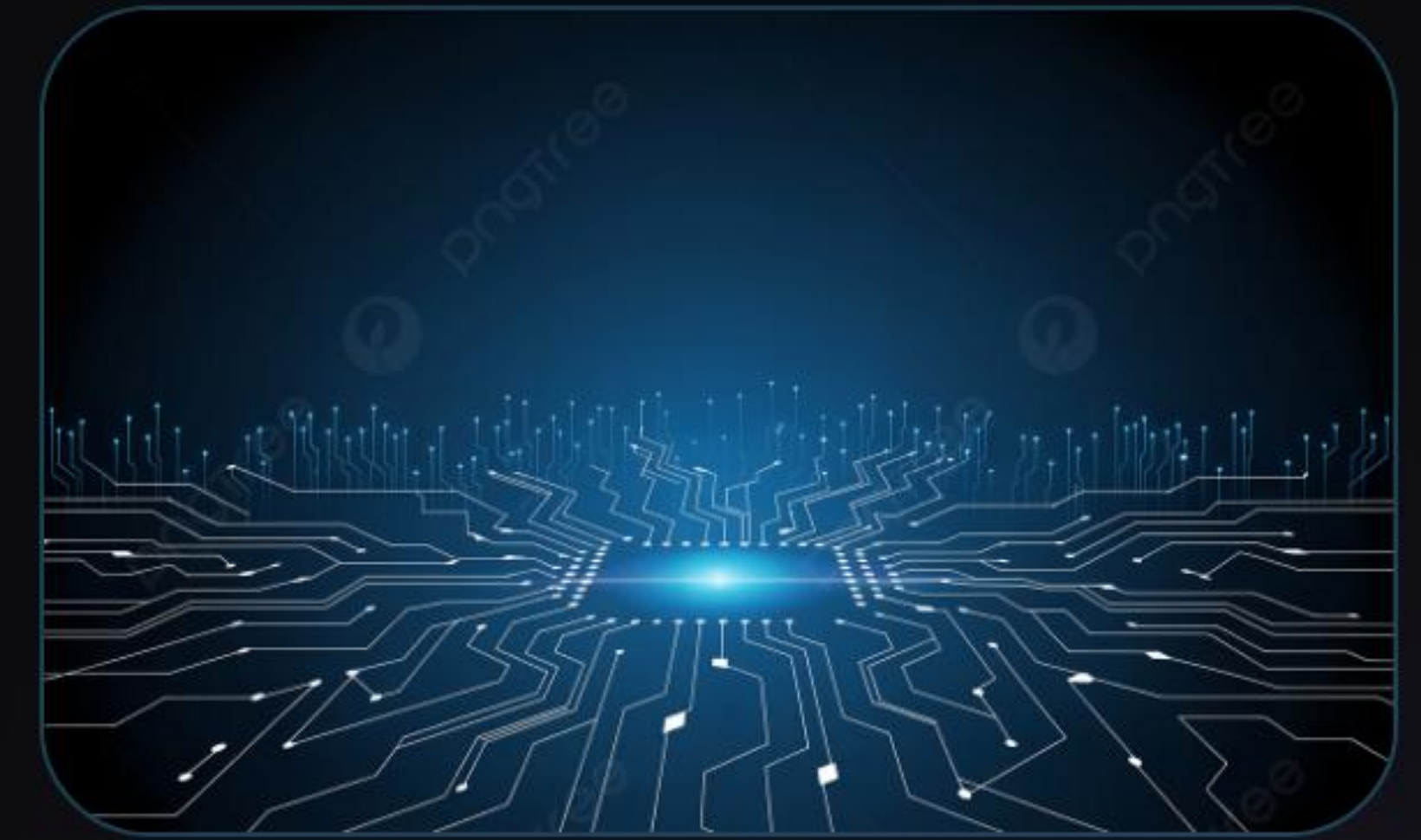
Education

Interactive tool for teaching students the fundamentals of shift ciphers.

Abstract game UI

Gamification

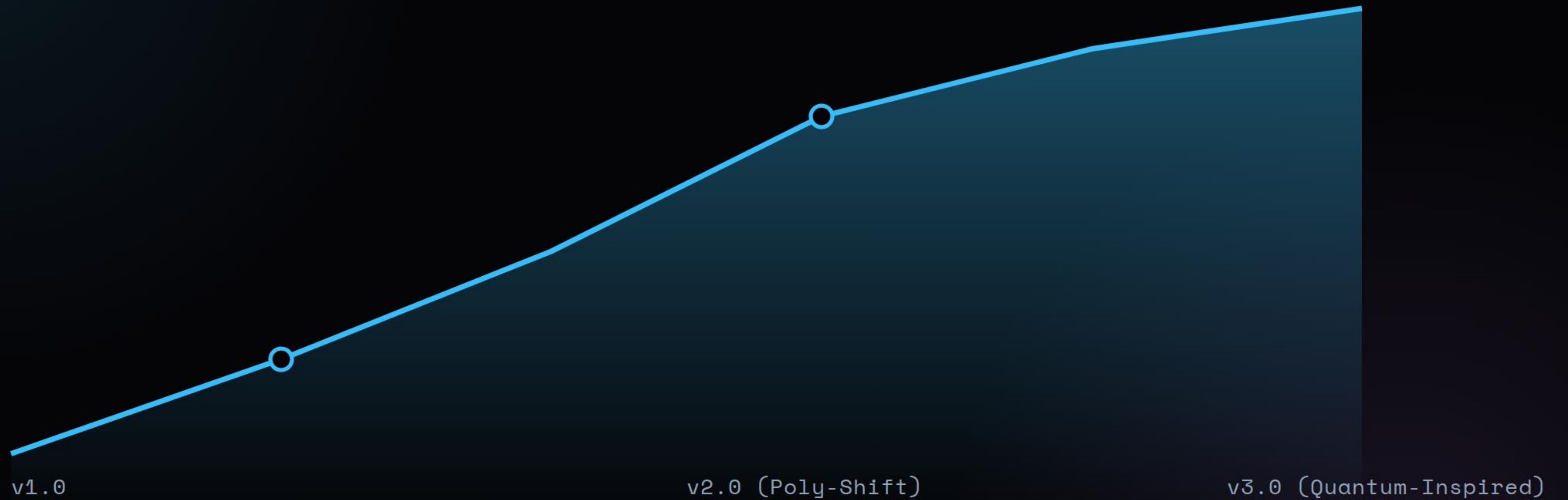
Integration into ARG (Alternate Reality Games) as a decryption minigame.



Dev Playground

Exploring high-performance CSS and JS animation techniques.

SCALABILITY POTENTIAL



Projected Complexity vs. Development Iteration

QUESTIONS?

Explore the source code on GitHub

`avenchaos43654-droid/WC5-03`



IMAGE SOURCES



https://png.pngtree.com/background/20250520/original/pngtree-digital-circuit-horizon-with-glowing-core-and-blue-background-picture-image_16570652.jpg

Source: [pngtree.com](https://png.pngtree.com)



https://static.vecteezy.com/system/resources/thumbnails/076/476/996/small_2x/futuristic-glassmorphism-ui-element-with-square-transparent-frame-and-liquid-glassy-texture-blue-neon-light-effect-on-3d-glass-button-box-with-round-edges-modern-app-interface-design-template-vector.jpg

Source: www.vecteezy.com



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